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(54) **INK JET RECORDING APPARATUS**

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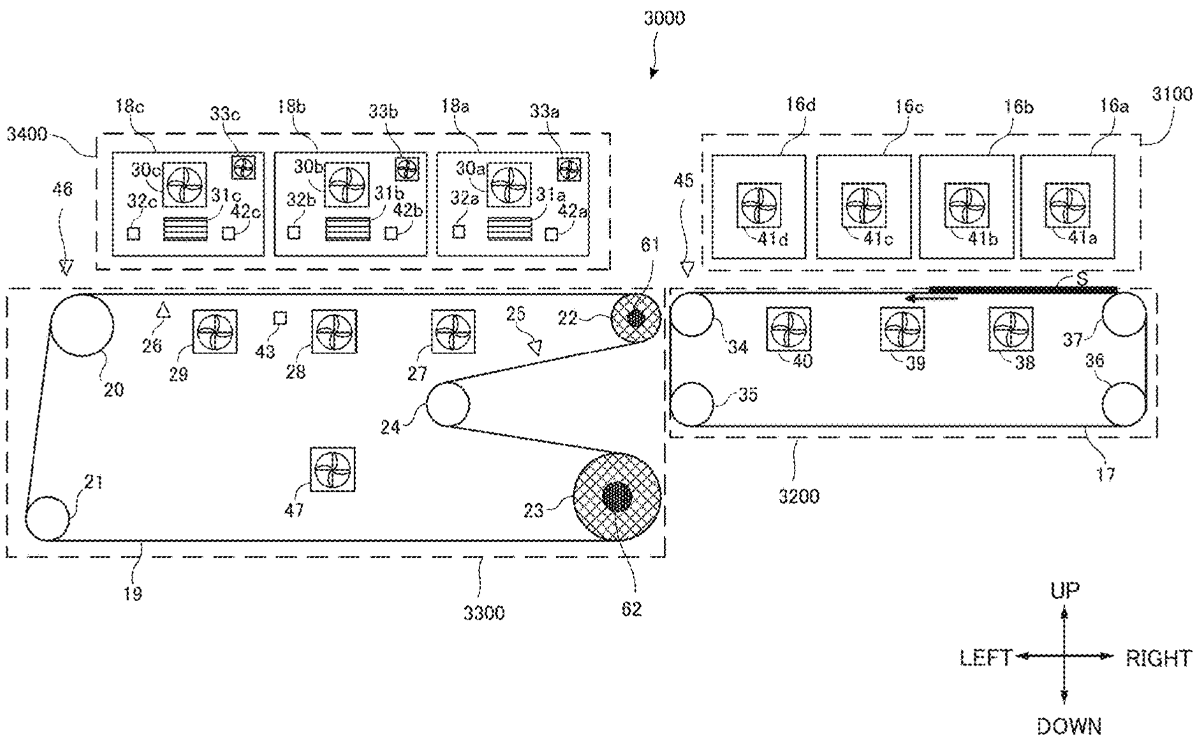
(57) **ABSTRACT**

An ink jet recording apparatus includes a conveyance portion, a drying portion including a heater and a blowing fan, a casing, a temperature sensor, an exhaust fan configured to exhaust air in the casing, a door provided to be opened and closed on the casing, a lock mechanism configured to lock the door to the casing in a closed state, and a control portion. In a case where the control portion receives an instruction to unlock the door locked by the lock mechanism, the control portion is configured to stop heating of the air by the heater, start an operation of the exhaust fan when the exhaust fan is stopped, continue the operation of the exhaust fan when the exhaust fan is in operation, and unlock the door locked by the lock mechanism after the temperature in the casing reaches a predetermined temperature or lower.



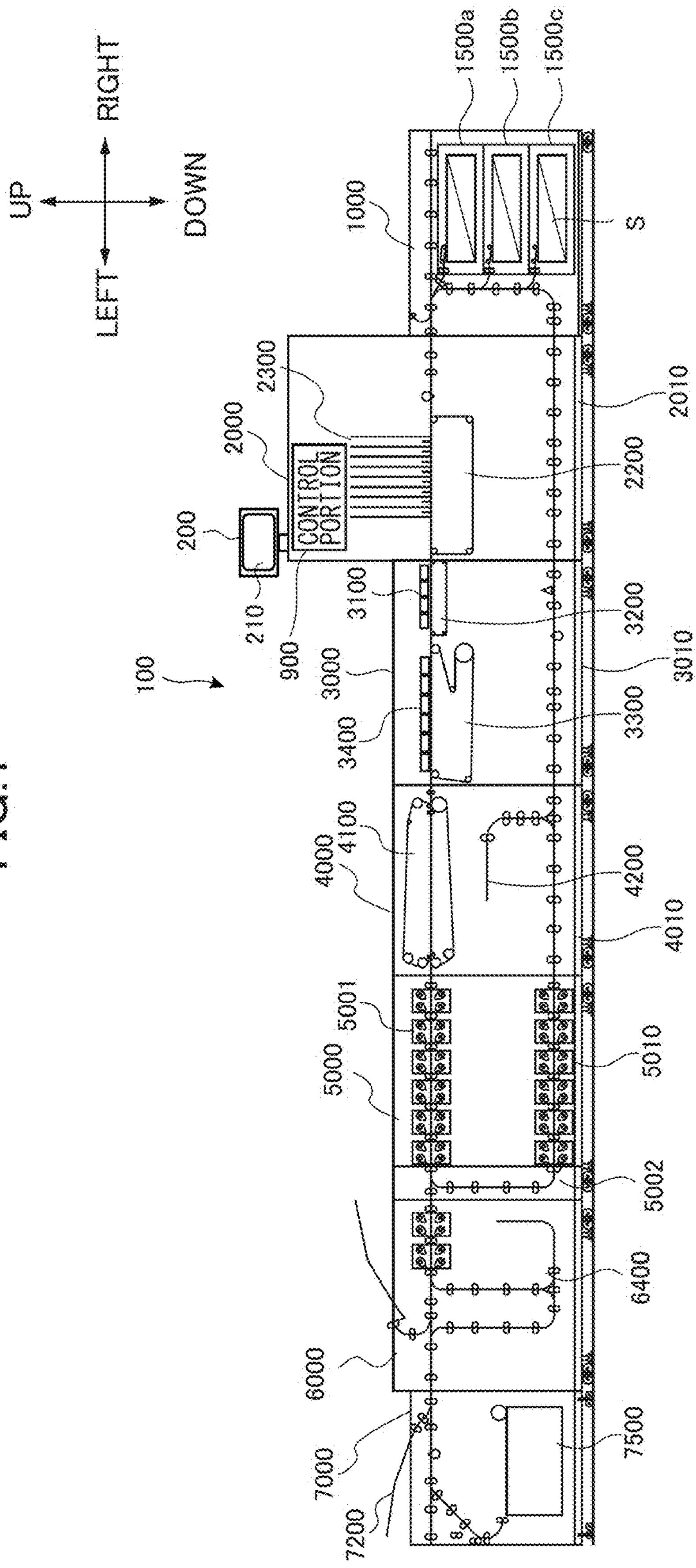

UP
DOWN
LEFT
RIGHT
100
200
210
900
1000
1500a
1500b
1500c
S
2000
CONTROL PORTION
2010
2200
2300
3000
3010
3100
3200
3300
3400
4000
4010
4100
4200
5000
5001
5002
5010
6000
6400
7000
7200
7500

FIG. 1

FIG.2A

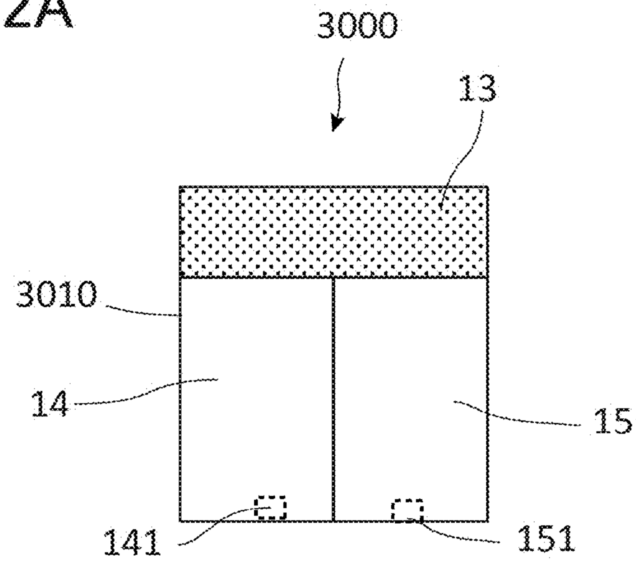


FIG.2B

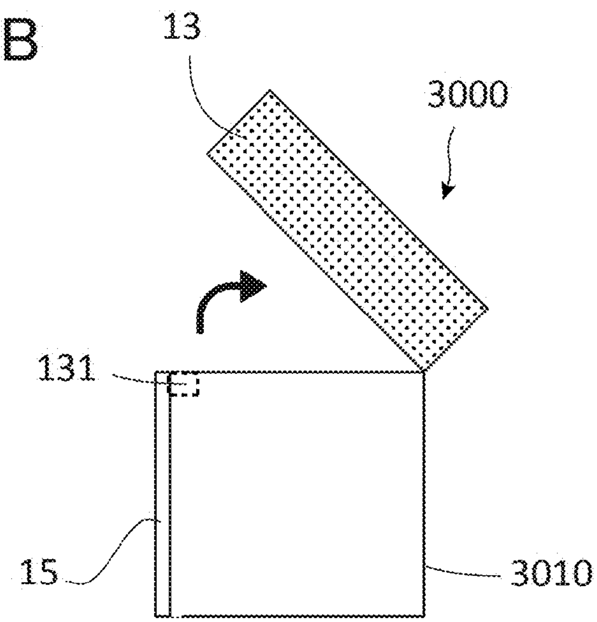


FIG.2C

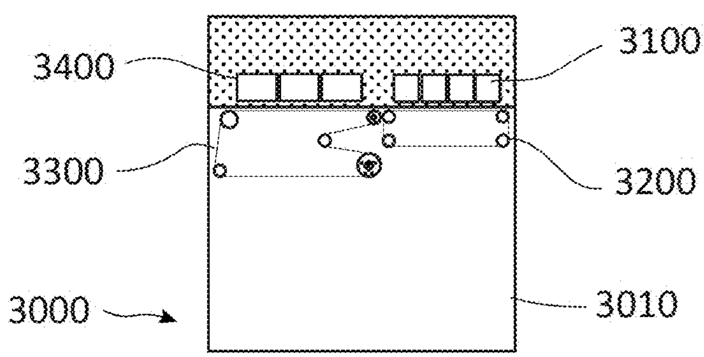


FIG.3

3000

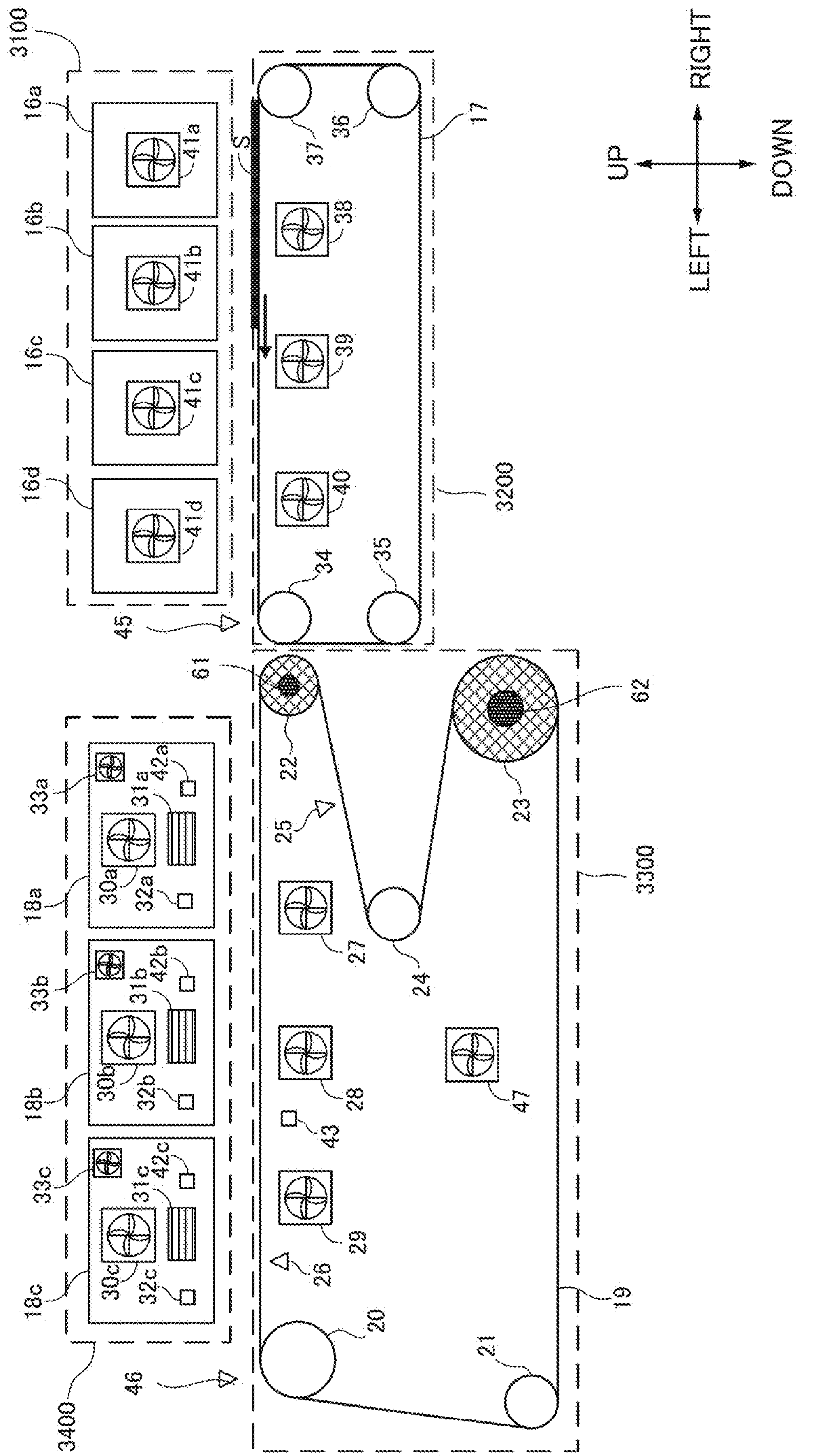

3100
16a
16b
16c
16d
41a
41b
41c
41d
S
17
36
37
38
39
40
34
35
3200
45
UP
DOWN
LEFT
RIGHT
61
62
22
23
25
24
27
28
47
43
29
26
20
21
19
3300
18a
18b
18c
33a
33b
33c
30a
30b
30c
31a
31b
31c
32a
32b
32c
42a
42b
42c
3400
46

FIG.4

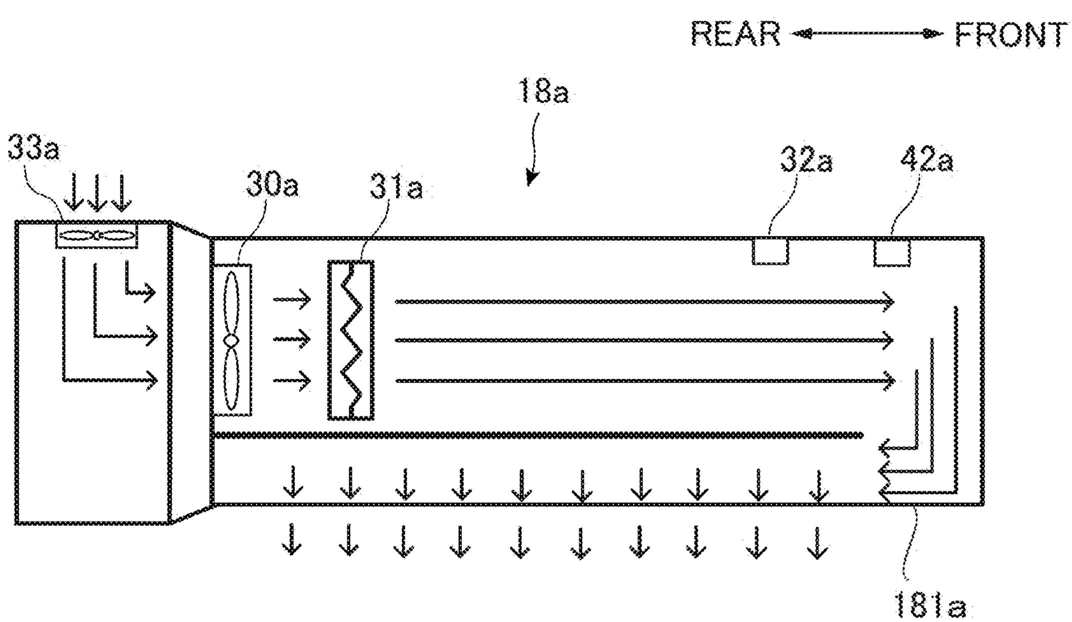


FIG.5

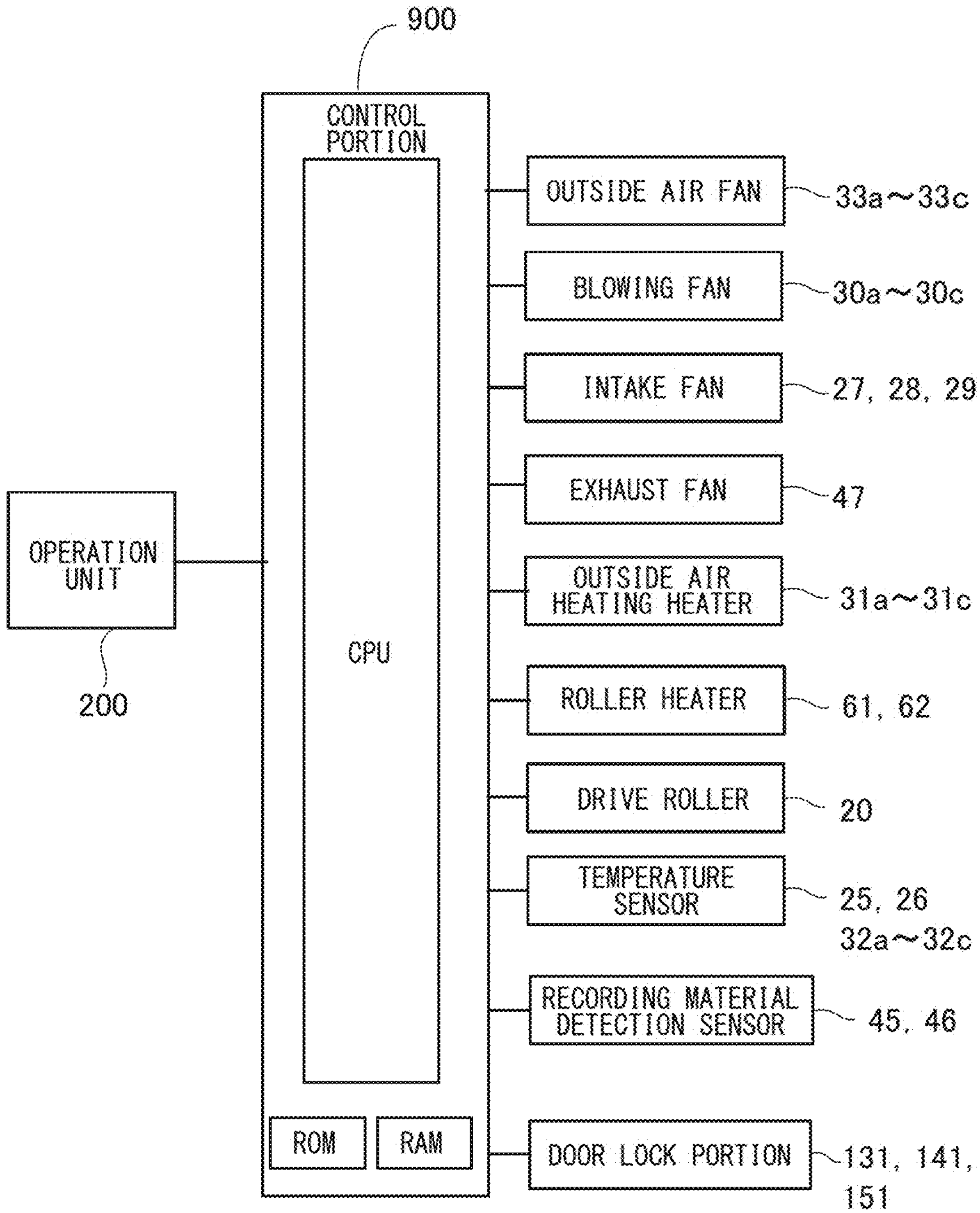

900
CONTROL PORTION
CPU
ROM
RAM
OPERATION UNIT
200
OUTSIDE AIR FAN
33a～33c
BLOWING FAN
30a～30c
INTAKE FAN
27, 28, 29
EXHAUST FAN
47
OUTSIDE AIR HEATING HEATER
31a～31c
ROLLER HEATER
61, 62
DRIVE ROLLER
20
TEMPERATURE SENSOR
25, 26
32a～32c
RECORDING MATERIAL DETECTION SENSOR
45, 46
DOOR LOCK PORTION
131, 141, 151

FIG.6

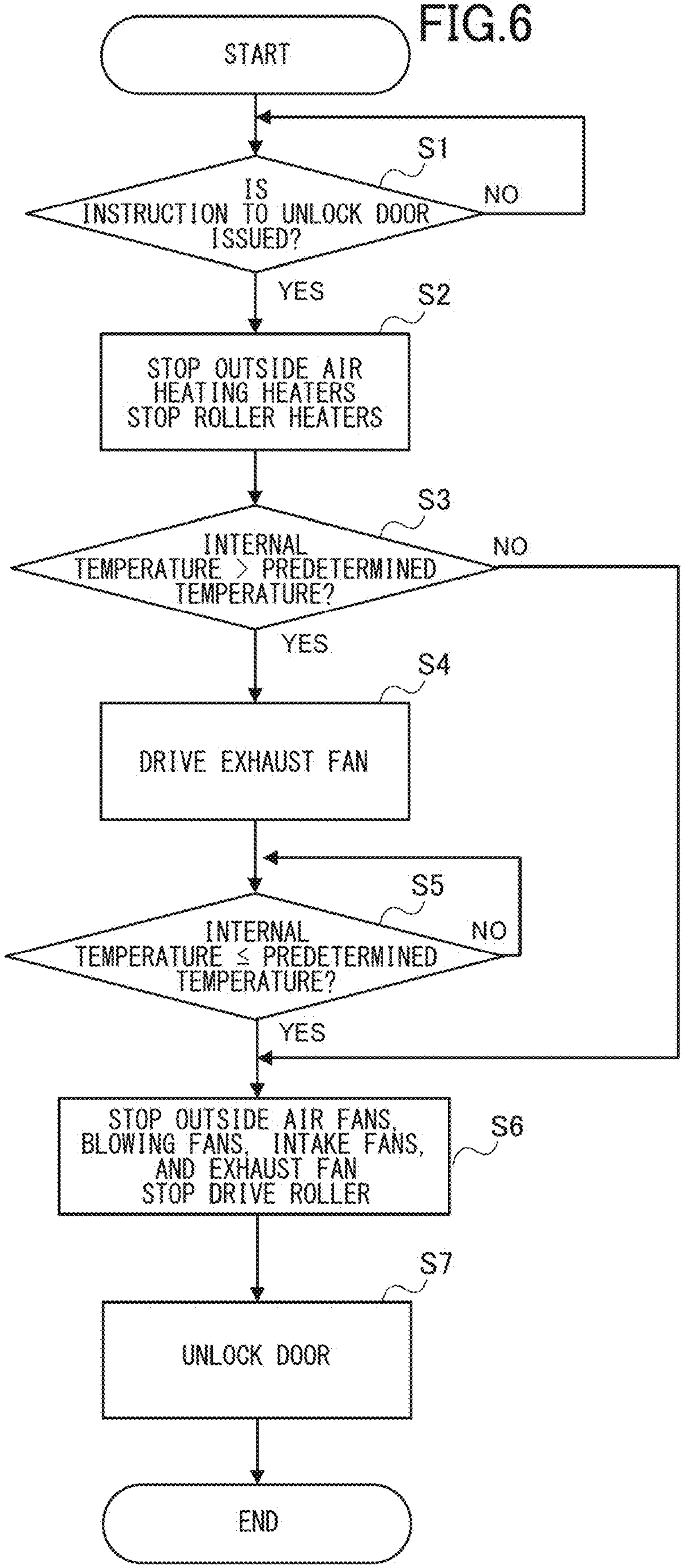
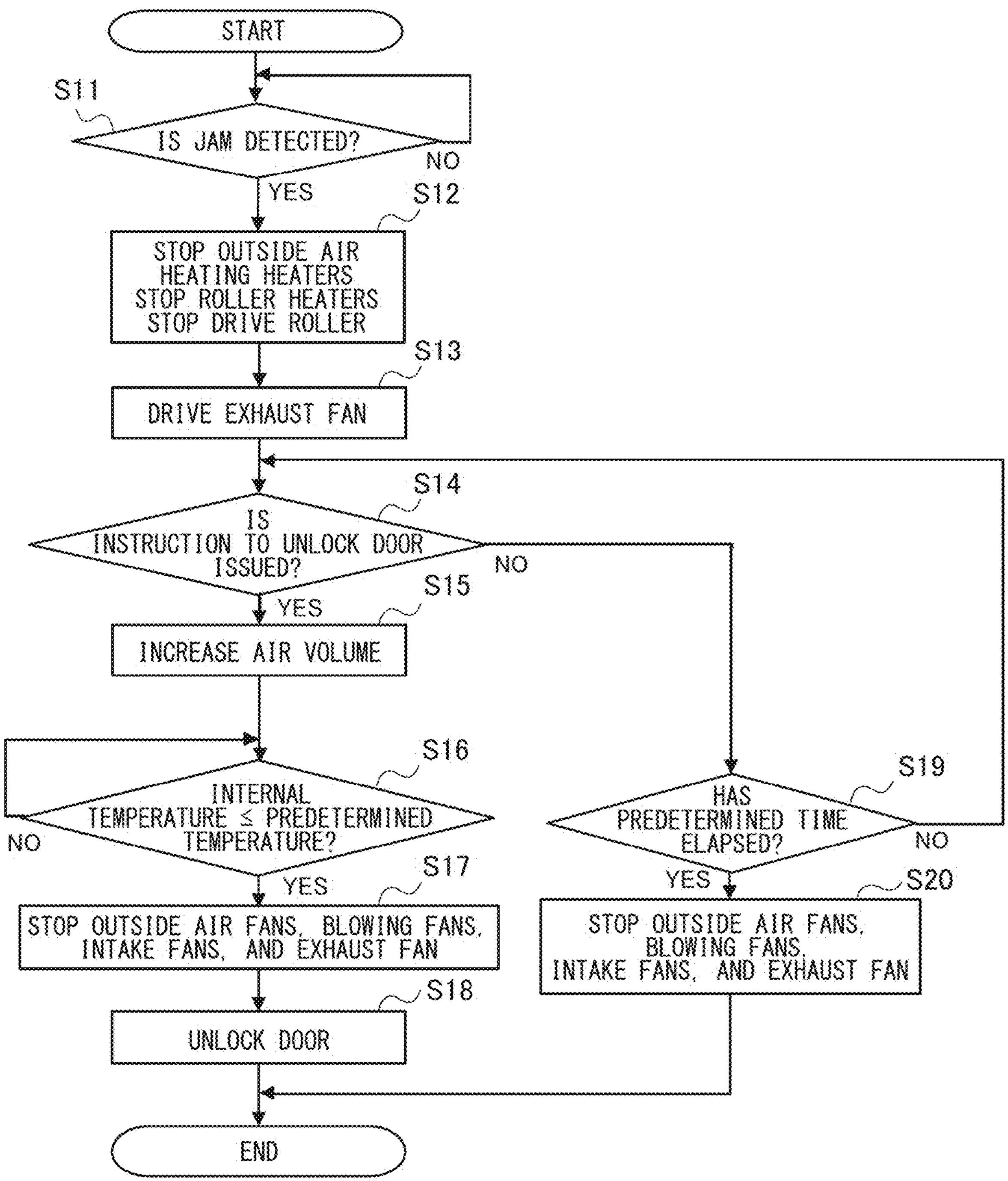

START
S1
IS INSTRUCTION TO UNLOCK DOOR ISSUED?
NO
YES
S2
STOP OUTSIDE AIR HEATING HEATERS
STOP ROLLER HEATERS
S3
INTERNAL TEMPERATURE > PREDETERMINED TEMPERATURE?
NO
YES
S4
DRIVE EXHAUST FAN
S5
INTERNAL TEMPERATURE ≤ PREDETERMINED TEMPERATURE?
NO
YES
STOP OUTSIDE AIR FANS, BLOWING FANS, INTAKE FANS, AND EXHAUST FAN
STOP DRIVE ROLLER
S6
S7
UNLOCK DOOR
END

FIG.7


START
S11
IS JAM DETECTED?
NO
YES
S12
STOP OUTSIDE AIR HEATING HEATERS
STOP ROLLER HEATERS
STOP DRIVE ROLLER
S13
DRIVE EXHAUST FAN
S14
IS INSTRUCTION TO UNLOCK DOOR ISSUED?
NO
YES
S15
INCREASE AIR VOLUME
S16
INTERNAL TEMPERATURE ≤ PREDETERMINED TEMPERATURE?
NO
YES
S17
STOP OUTSIDE AIR FANS, BLOWING FANS, INTAKE FANS, AND EXHAUST FAN
S18
UNLOCK DOOR
S19
HAS PREDETERMINED TIME ELAPSED?
NO
YES
S20
STOP OUTSIDE AIR FANS, BLOWING FANS, INTAKE FANS, AND EXHAUST FAN
END

INK JET RECORDING APPARATUS

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an ink jet recording apparatus that forms an image by ink.

Description of the Related Art

[0002] In the related art, an ink jet recording apparatus that forms an image on a recording material by ink has been proposed. The ink jet recording apparatus described in Japanese Patent Application Laid-Open Publication No. 2023-60712 includes a drying portion that blows high-temperature air to a recording material on which an image is formed by ink to dry the recording material (Japanese Patent Application Laid-Open Publication No. 2023-60712). In the drying portion, the recording material is dried by blowing high-temperature air to the recording material conveyed on a rotating conveyor belt.

[0003] In the ink jet recording apparatus, a door that can be opened and closed is disposed in a casing so that a user can perform maintenance and the user can remove a clogged recording material in a case where a so-called jam occurs in which the recording material is clogged in the middle of being conveyed. However, the door is locked not to be opened unless the user issues an unlocking instruction to open the door.

SUMMARY OF THE INVENTION

[0004] According to a first aspect of the present invention, an ink jet recording apparatus forming an image on a recording material by jetting an ink, the apparatus includes a conveyance portion configured to convey the recording material on which the image is formed by the ink, a drying portion including a heater and a blowing fan blowing air toward the conveyance portion and configured to dry the ink by blowing air heated by the heater to the recording material conveyed by the conveyance portion, a casing configured to accommodate the conveyance portion, a temperature sensor configured to detect a temperature in the casing, an exhaust fan configured to exhaust air in the casing to an outside of the casing and to be controlled to an operation state and a stop state based on a detection result of the temperature sensor, a door provided to be opened and closed on the casing to allow access to the conveyance portion, a lock mechanism configured to lock the door to the casing in a closed state, and a control portion configured to control the drying portion, the lock mechanism, and the exhaust fan. In a case where the control portion receives an instruction to unlock the door locked by the lock mechanism, the control portion is configured to stop heating of the air by the heater, start an operation of the exhaust fan when the exhaust fan is stopped, continue the operation of the exhaust fan when the exhaust fan is in operation, and unlock the door locked by the lock mechanism after the temperature in the casing reaches a predetermined temperature or lower.

[0005] According to a second aspect of the present invention, an ink jet recording apparatus forming an image on a recording material by jetting an ink, the apparatus includes a conveyance portion configured to convey the recording material on which the image is formed by the ink, a detection unit configured to detect the recording material conveyed by the conveyance portion, a drying portion including a heater and a blowing fan blowing air toward the conveyance portion and configured to dry the ink by blowing air heated by the heater to the recording material conveyed by the conveyance portion, a casing configured to accommodate the conveyance portion, a temperature sensor configured to detect a temperature in the casing, an exhaust fan configured to exhaust air in the casing to an outside of the casing and to be controlled to an operation state and a stop state based on a detection result of the temperature sensor, a door that is provided to be opened and closed on the casing to allow access to the conveyance portion, a lock mechanism configured to lock the door to the casing in a closed state, and a control portion configured to control the drying portion, the lock mechanism, and the exhaust fan. In a case where an error occurs in a detection result of the detection unit, the control portion is configured to stop heating of the air by the heater, and start the operation of the exhaust fan when the exhaust fan is stopped. When the control portion receives an instruction to unlock the door locked by the lock mechanism after the error occurs, the control portion is configured to unlock the door locked by the lock mechanism after the temperature in the casing reaches a predetermined temperature or lower.

[0006] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a schematic view showing an ink jet recording apparatus.

[0008] FIG. **2**A is a front view showing a drying module.

[0009] FIG. **2**B is a right side view showing the drying module.

[0010] FIG. **2**C is a schematic view.

[0011] FIG. **3** is a schematic view showing a configuration of the drying module.

[0012] FIG. **4** is a cross-sectional view showing a hot air blowing unit.

[0013] FIG. **5** is a control block diagram showing a control system of the drying module.

[0014] FIG. **6** is a flowchart showing door unlocking processing of a first embodiment.

[0015] FIG. **7** is a flowchart showing door unlocking processing according to a second embodiment.

DESCRIPTION OF THE EMBODIMENTS

First Embodiment

Ink Jet Recording Apparatus

[0016] Hereinafter, embodiments of the present invention will be described with reference to drawings. First, an ink jet recording apparatus of the present embodiment will be described with reference to FIG. **1**. An ink jet recording apparatus **100** of the present embodiment is a so-called sheet-fed ink jet recording apparatus that forms an image on a recording material by using an ink and a reaction solution based on image information input from an external apparatus such as a computer or a document reading apparatus. The recording material may be, for example, a recording material capable of receiving ink, such as paper such as plain paper and cardboard, plastic film such as a recording material for overhead projectors, specially shaped recording materials such as envelopes and index paper, and cloth.

[0017] In the present specification, a side on which the user stands when operating the ink jet recording apparatus **100** is referred to as a "front (or front side)", and an opposite side thereof is referred to as a "rear (or rear side)". In addition, the left side when viewed from the front is referred to as "left", and the right side when viewed from the front is referred to as "right". FIG. **1** shows a case where the ink jet recording apparatus **100** is viewed from the front.

[0018] As shown in FIG. **1**, the ink jet recording apparatus **100** includes a feeding module **1000**, a printing module **2000**, a drying module **3000**, a fixing module **4000**, a cooling module **5000**, a reverse module **6000**, and a loading module **7000**. A recording material S supplied from the feeding module **1000** is subjected to various types of processing while being conveyed along the conveyance path in each module, and is finally discharged to the loading module **7000**.

[0019] The feeding module **1000**, the printing module **2000**, the drying module **3000**, the fixing module **4000**, the cooling module **5000**, the reverse module **6000**, and the loading module **7000** may each have a separate casing, and the ink jet recording apparatus **100** may be configured by connecting the casings. For example, a casing **2010** of the printing module **2000** and a casing **3010** of the drying module **3000**, the casing **3010** and a casing **4010** of the fixing module **4000**, and the casing **4010** and a casing **5010** of the cooling module **5000** are connected to each other so that the recording material S can be transferred between the respective casings. In the present embodiment, the feeding module **1000**, the printing module **2000**, the drying module **3000**, the fixing module **4000**, the cooling module **5000**, the reverse module **6000**, and the loading module **7000** may be disposed in one casing.

[0020] The feeding module **1000** includes cassettes **1500*a***, **1500*b***, and **1500*c*** that accommodate the recording material S, and the cassettes **1500*a*** to **1500*c*** are provided to be pulled out to the front side in order to accommodate the recording material S. The recording material S is fed one by one by a separation belt and a conveyance roller in each of the cassettes **1500*a*** to **1500*c***, and is conveyed to the printing module **2000**.

[0021] The printing module **2000** includes a pre-imaging conveyance unit (not shown), a printing belt unit **2200**, and a recording portion **2300**. The recording material S conveyed from the feeding module **1000** is conveyed to the printing belt unit **2200** after the inclination or position of the recording material S is corrected by the pre-imaging conveyance unit. The recording portion **2300** is disposed at a position facing the printing belt unit **2200** with respect to the conveyance path of the recording material S. The recording portion **2300** forms an image by jetting ink onto the recording material S by a plurality of recording heads from above the recording material S conveyed by the belt. The recording material S is suctioned and conveyed by the printing belt unit **2200** to ensure clearance with the recording heads.

[0022] In addition, the printing module **2000** includes a control portion **900**, and the control portion **900** controls the entire ink jet recording apparatus **100**. The control portion **900** has, for example, a central processing unit (CPU), a read only memory (ROM), and a random access memory (RAM) (see FIG. **5**). Various programs such as an image forming job and various types of data such as image data are stored in the ROM or the RAM. For example, the CPU can operate the ink jet recording apparatus **100** to perform image formation by executing an image forming program stored in the ROM or the RAM. The RAM can temporarily store, for example, a calculation processing result obtained by executing various programs.

[0023] Further, the printing module **2000** is provided with an operation unit **200**. The operation unit **200** includes a liquid crystal display unit **210** that can display various types of information in addition to physically provided switches. The liquid crystal display unit **210** shows various programs, various types of data, or various displays such as the end of an image forming job or the occurrence of a jam caused by the recording material S being clogged in the middle of being conveyed. The operation unit **200** may be a touch panel that can receive, for example, a start input of various programs such as an image forming job, inputs of various types of data, or an instruction to unlock the door described below in response to a touch operation of the user. The start input of the various programs, the inputs of various types of data, an instruction to unlock the door, and the like are not limited to be input from the operation unit **200**, and may be input from an external apparatus such as a computer.

[0024] The image forming job is a series of operations from the start of image formation to the completion of the image forming operation based on the image signal for forming an image on the recording material S. That is, the image forming job is a series of operations from the start of a preliminary operation (so-called previous rotation) required for performing image formation to the completion of a preliminary operation (so-called subsequent rotation) required for completing the image formation through an image forming step. Specifically, the image forming job refers to a period from a previous rotation after receiving the image signal (preparatory operation before image formation) to a subsequent rotation (operation after image formation), and includes an image formation period and a paper gap.

[0025] The drying module **3000** has an outside air blowing unit **3100**, a decoupling portion **3200**, a drying belt unit **3300**, and a hot air blowing unit **3400**. The drying module **3000** reduces the liquid component of the ink applied to the recording material S in order to increase the fixing property of the ink to the recording material S by the subsequent fixing module **4000**.

[0026] The recording material S on which an image is formed is conveyed to the decoupling portion **3200** disposed in the drying module **3000**. In the decoupling portion **3200**, a frictional force is generated between the recording material S and the belt by the wind pressure of the air blown from the outside air blowing unit **3100** disposed above the decoupling portion **3200**, and the recording material S is conveyed by the belt. In this way, the recording material S placed on the belt is conveyed by the frictional force, and thus the recording material S is prevented from deviating during the conveyance of the recording material S over the printing belt unit **2200** and the decoupling portion **3200**.

[0027] The recording material S conveyed from the decoupling portion **3200** is conveyed by the drying belt unit **3300** serving as a conveyance portion. The hot air blowing unit **3400** is disposed above the drying belt unit **3300**, and hot air heated by a heater to a high temperature is blown from the hot air blowing unit **3400** serving as a drying portion to the recording material S conveyed from the drying belt unit **3300**. In this manner, the ink applied to the recording material S is dried.

[0028] The fixing module **4000** includes a fixing belt unit **4100**. The fixing belt unit **4100** fixes the ink to the recording material S by causing the recording material S conveyed from the drying module **3000** to pass between the heated upper belt unit and the lower belt unit.

[0029] The cooling module **5000** has a plurality of cooling portions **5001** and cools the recording material S having a high temperature conveyed from the fixing module **4000** by the cooling portions **5001**. Although the cooling portion **5001** is not shown, the cooling portion **5001** takes in air (outside air) into a cooling box by a fan from the outside of the casing **5010** to increase the pressure in the cooling box, and applies air, which is ejected from the cooling box through a nozzle by the pressure, to the recording material S to cool the recording material S. The cooling portions **5001** are disposed on both sides of the recording material S in the conveyance path of the recording material S, and cools both sides of the recording material S.

[0030] In addition, the cooling module **5000** is provided with a conveyance path switching portion **5002**. The conveyance path switching portion **5002** switches the conveyance path of the recording material S according to whether the recording material S is conveyed to the reverse module **6000** or whether the recording material S is conveyed to the double-sided conveyance path for double-sided printing, in which images are formed on both sides of the recording material S.

[0031] The reverse module **6000** includes a reverse portion **6400**. The reverse portion **6400** reverses the front and back of the recording material S to be conveyed and changes the front and back orientation of the recording material S when discharging the recording material S to the supporting module **7000**. The supporting module **7000** has a top tray **7200** and a supporting portion **7500**, and supports the recording material S conveyed from the reverse module **6000**.

[0032] In a case of double-sided printing, the recording material S is conveyed to a conveyance path below the cooling module **5000** by the conveyance path switching portion **5002**. Thereafter, the recording material S is returned to the printing module **2000** through the double-sided conveyance path of the fixing module **4000**, the drying module **3000**, the printing module **2000**, and the feeding module **1000**. The double-sided conveyance portion of the fixing module **4000** is provided with a reverse portion **4200** that reverses the front and back of the recording material S. The recording material S returned to the printing module **2000** has an image formed by ink on the other side on which an image is not formed, and is discharged to the supporting module **7000** through the drying module **3000**, the fixing module **4000**, the cooling module **5000**, and the reverse module **6000**.

Drying Module

[0033] Next, the drying module **3000** will be described with reference to FIGS. **2**A to **4**. As shown in FIG. **2**A, a left front door **14**, a right front door **15**, and an upper door **13** are disposed to be opened and closed on the casing **3010** of the drying module **3000**. The left front door **14** is provided to be pivotable about a pivot shaft provided on a left side of the casing **3010** serving as a pivot axis, and the right front door **15** is provided to be pivotable about a pivot shaft provided on a right side of the casing **3010** serving as a pivot axis. That is, the left front door **14** and the right front door **15** are double doors that are opened and closed in a direction intersecting the casing **3010** in the up-down direction.

[0034] As shown in FIG. **2**B, the upper door **13** is provided to be pivotable about a pivot shaft provided on the rear side of the casing **3010** serves as a pivot axis. That is, by supporting the upper door **13** to be pivotable on the rear side, the front side of the upper door **13** can be opened upward so that the user can easily work from the front. As shown in FIG. **2**C, the outside air blowing unit **3100** and the hot air blowing unit **3400** are attached to the upper door **13**, and the outside air blowing unit **3100** and the hot air blowing unit **3400** are moved integrally with the upper door **13**. In this case, the decoupling portion **3200** and the drying belt unit **3300** remain on the casing **3010** side. In the present specification, the left front door **14** and the right front door **15** to which the unit is not attached unlike the upper door **13**, and the upper door **13** to which the outside air blowing unit **3100** and the hot air blowing unit **3400** are attached are referred to as "door". Among these doors, the left front door **14** and the upper door **13** are doors that are provided to be opened and closed in the casing **3010** in order to allow access to the drying belt unit **3300** accommodated in the casing **3010**.

[0035] The user can perform maintenance on the outside air blowing unit **3100**, the decoupling portion **3200**, the drying belt unit **3300**, and the hot air blowing unit **3400** in the casing **3010** by opening the upper door **13**, the left front door **14**, and the right front door **15**. In addition, the user can remove the recording material S from the decoupling portion **3200** or the drying belt unit **3300** by opening the upper door **13**, the left front door **14**, and the right front door **15** when a jam occurs.

[0036] The drying module **3000** is provided with door lock portions (**131**, **141**, and **151**) serving as lock mechanisms that can lock the upper door **13**, the left front door **14**, and the right front door **15**, respectively. When the main power supply of the ink jet recording apparatus **100** is turned on, the door lock portion **131**, the door lock portion **141**, and the door lock portion **151** lock the upper door **13**, the left front door **14**, and the right front door **15**, respectively, in a closed state not to be opened. The locked state of each door by the door lock portions (**131**, **141**, and **151**) is unlocked by the door unlocking based on the unlocking signal transmitted from the control portion **900** in response to the user's instruction to unlock the door from the operation unit **200**. The door lock portions (**131**, **141**, and **151**) are not shown, but each of the door lock portions has, for example, a locking claw that is provided to be pivotable and a motor or the like that causes the locking claw to pivot. The control portion **900** controls the motor to pivot the locking claw.

Decoupling Portion

[0037] The outside air blowing unit **3100**, the decoupling portion **3200**, the drying belt unit **3300**, and the hot air blowing unit **3400** of the drying module **3000** will be described in detail. As shown in FIG. **3**, the decoupling portion **3200** includes an endless conveyor belt **17**, a drive roller **34**, driven rollers **35**, **36**, and **37**, and fans **38**, **39**, and **40**. The conveyor belt **17** is stretched around the drive roller **34** and the driven rollers **35**, **36**, and **37**, and is rotated as the drive roller **34** is driven by a motor (not shown). A large number of minute holes are formed in the conveyor belt **17**, and the fans **38**, **39**, and **40** are disposed on the inner peripheral side of the conveyor belt **17**. In addition, a recording material detection sensor **45** that detects the recording material S conveyed by the conveyor belt **17** is disposed at a position in the conveyor belt **17** facing the drive roller **34**.

Outside Air Blowing Unit

[0038] The outside air blowing unit **3100** includes four identical air units (**16***a*, **16***b*, **16***c*, and **16***d*), and the air units (**16***a* to **16***d*) are disposed side by side in the conveyance direction of the recording material S. The air units (**16***a* to **16***d*) have fans (**41***a*, **41***b*, **41***c*, and **41***d*), and the fans (**41***a* to **41***d*) blow air (outside air) taken in from the outside of the casing **3010** (see FIG. **1**) toward the recording material conveyance surface of the opposing conveyor belt **17**. As described above, since the minute holes are formed in the conveyor belt **17**, the air blown by the fans (**41***a* to **41***d*) escapes from the recording material conveyance surface to the inner peripheral side and is taken in by the fans **38**, **39**, and **40** disposed on the inner peripheral side of the conveyor belt **17**. In a state in which the recording material S is pressed against the conveyor belt **17** by blowing air, the inner peripheral surface side of the conveyor belt **17** is in a negative pressure state compared to the recording material conveyance surface side due to the intake of the fans **38**, **39**, and **40**. Therefore, the recording material S is conveyed in a state of being stuck to the conveyor belt **17**. In this way, the recording material S is stuck to the conveyor belt **17**, and thus the recording material S can be stably conveyed.

[0039] In the outside air blowing unit **3100**, the outside air taken in by the fans (**41***a* to **41***d*) is blown without being heated, and the moisture of the ink applied to the recording material S is not dried. This is to prevent the recording material S from being deformed in a wave shape called cockling in a case where image formation by the printing module **2000** and drying of the ink by the drying module **3000** are performed at the same time on the recording material S being conveyed over the printing module **2000** and the drying module **3000**.

Drying Belt Unit

[0040] The drying belt unit **3300** includes an endless heating belt **19**, a drive roller **20**, driven rollers **21** and **24**, heating rollers **22** and **23**, temperature sensors **25** and **26**, intake fans **27**, **28**, and **29**, and an exhaust fan **47**. The heating belt **19** is stretched over the drive roller **20**, the driven rollers **21** and **24**, and the heating rollers **22** and **23**, and is rotated as the drive roller **20** is driven by a motor (not shown). A large number of minute holes are formed in the heating belt **19**, and the intake fans **27**, **28**, and **29**, the temperature sensors **25** and **26**, a pressure sensor **43**, and the exhaust fan **47** are disposed on the inner peripheral side of the heating belt **19**.

[0041] The heating belt **19** is heated by the heating rollers **22** and **23** serving as heating portions. The heating rollers **22** and **23** are provided with roller heaters **61** and **62** therein. The roller heaters **61** and **62** are, for example, halogen heaters, and heat is generated by power supply from a control board (not shown). Here, the amount of power supplied to the roller heaters **61** and **62** is determined based on the temperature sensors **25** and **26** that measure the surface temperature of the heating belt **19**. As the temperature sensors **25** and **26**, for example, an infrared temperature sensor that can detect the temperature of the heating belt **19** in a non-contact manner is used. In addition, a recording material detection sensor **46** serving as a detection unit that detects the recording material S conveyed by the heating belt **19** is disposed at a position facing the drive roller **20** in the heating belt **19**.

[0042] In the recording material S conveyed to the heating belt **19**, high-temperature hot air is blown from the hot air blowing unit **3400** disposed above the heating belt **19** to dry the moisture of the ink. The hot air blowing unit **3400** will be described below.

[0043] As described above, since the minute holes are formed in the heating belt **19**, the hot air blown from the hot air blowing unit **3400** escapes from the recording material conveyance surface to the inner peripheral side and is taken in by the intake fans **27**, **28**, and **29** disposed on the inner peripheral side of the conveyor belt **17**. When the recording material S is pressed against the heating belt **19** by blowing hot air, the inner peripheral surface side of the heating belt **19** is in a negative pressure state compared to the recording material conveyance surface side due to the intake of the intake fans **27**, **28**, and **29**. Therefore, the recording material S is conveyed in a state of being stuck to the heating belt **19**. By sticking the recording material S to the heating belt **19** in this way, the recording material S can be stably conveyed. In addition, since the recording material S in the middle of being conveyed is heated by the heating belt **19**, the drying performance of the ink is improved.

[0044] The intake fans **27**, **28**, and **29** are controlled such that the detection value of the pressure sensor **43** that measures the static pressure is a target value. As the intake fans **27**, **28**, and **29**, for example, a type of fan that controls drive power by a duty ratio of an input PWM signal is used. In this case, the duty ratio of the PWM signal is fed back, and the drive power of the intake fans **27**, **28**, and **29** is controlled. As the pressure sensor **43**, a sensor of a type that outputs a difference value between the static pressure at the location where the sensor is disposed and the atmospheric pressure. The target value of the difference value is, for example, "−1000 pascals".

[0045] The exhaust fan **47** is provided to discharge air in the casing **3010** (in the casing) to the outside (outside the casing). During the image forming job, the exhaust fan **47** is controlled to be turned on and off such that the temperature in the casing **3010** is kept in a predetermined range (for example, 80° C. to 90° C.) higher than the normal temperature based on the detection results of the temperature sensors **25** and **26**. In this manner, the drying performance of the ink during image formation is maintained. In the present embodiment, an average of the detection results of the temperature sensors **25** and **26** may be used as the temperature of the casing **3010**, or the detection result of any one of the temperature sensors **25** and **26** may be used as the temperature of the casing **3010**.

Hot Air Blowing Unit

[0046] The hot air blowing unit **3400** includes three identical heat air units (**18***a*, **18***b*, and **18***c*). The heat air units (**18***a* to **18***c*) are disposed side by side in the conveyance direction of the recording material S in order to ensure the drying time of the ink and to achieve high drying performance of the ink. The heat air units (**18***a* to **18***c*) will be described below with the heat air unit **18***a* as an example.

[0047] As shown in FIG. **4**, the heat air unit **18***a* includes a blowing fan **30***a*, an outside air heating heater **31***a*, a temperature sensor **32***a*, an outside air fan **33***a*, a pressure sensor **42***a*, and a duct **181***a*. The outside air fan **33***a* takes air (outside air) in from the outside of the casing **3010** (see FIG. **1**). The blowing fan **30***a* is disposed at the intake port of the duct **181***a*, and air taken in by the outside air fan **33***a* is taken into the duct **181***a*. The outside air heating heater **31***a* heats the air taken into the duct **181***a* by the blowing fan **30***a*. The temperature sensor **32***a* detects the temperature of the air passing through the duct **181***a* after the air is heated by the outside air heating heater **31***a*. The temperature of the outside air heating heater **31***a* is controlled such that the detection result of the temperature sensor **32***a* is a target temperature. A plurality of exhaust holes are formed in the duct **181***a* on a side facing the drying belt unit **3300**. The air taken into the duct **181***a* by the blowing fan **30***a* is heated to a high-temperature hot air by the outside air heating heater **31***a* and is blown toward the recording material conveyance surface of the heating belt **19** through the exhaust holes.

[0048] The blowing fan **30***a* is controlled such that the detection value of the pressure sensor **42***a* that measures the static pressure is a target value. As the blowing fan **30***a*, for example, a type of fan that controls drive power by a duty ratio of an input PWM signal is used. In this case, the duty ratio of the PWM signal is fed back, and the drive power of the blowing fan **30***a* is controlled. The pressure sensor **42***a* is disposed in the flow path of the air passing through the duct **181***a* after heating by the outside air heating heater **31***a*, and outputs the difference value between the static pressure at the location where the pressure sensor **42***a* is disposed and the atmospheric pressure. The target value of the difference value is, for example, "500 pascals".

Control System of Drying Module

[0049] Next, a control system of the drying module **3000** by the control portion **900** will be described by using FIG. **5** while referring to FIGS. **1** to **3**. Here, the control system of the drying belt unit **3300** and the hot air blowing unit **3400** will be mainly described. In addition, various apparatuses included in each module other than the drying module **3000** are connected to the control portion **900**, but the illustration and description thereof are omitted here because the apparatuses are not the main point of the invention.

[0050] As shown in FIG. **5**, the outside air fans **33***a* to **33***c*, the blowing fans **30***a* to **30***c*, the intake fans **27**, **28**, and **29**, and the exhaust fan **47** are connected to the control portion **900**, and the operation of these fans can be controlled. The outside air heating heaters **31***a* to **31***c*, the roller heaters **61** and **62**, the temperature sensors **25** and **26**, and the temperature sensors **32***a* to **32***c* are connected to the control portion **900**. The control portion **900** can control the roller heaters **61** and **62** based on the detection results of the temperature sensors **25** and **26**, and can control the outside air heating heaters **31***a* to **31***c* based on the detection results of the temperature sensors **32***a* to **32***c*.

[0051] In addition, the recording material detection sensors **45** and **46** and the drive roller **20** are connected to the control portion **900**. The recording material detection sensors **45** and **46** are provided to detect a delay and an early arrival of the conveyance timing of the recording material S in the drying module **3000**. The control portion **900** increases the conveyance speed of the conveyor belt **17** and the heating belt **19** in a case where the conveyance timing of the recording material S is late, and decreases the conveyance speed of the conveyor belt **17** and the heating belt **19** in a case where the conveyance timing is early. In addition, in a case where an early arrival or delay for more than a predetermined time occurs at the recording material detection sensors **45** and **46**, the conveyance control of the recording material S is not operating properly, and thus the image forming operation is stopped because an error is determined.

[0052] In the present embodiment, the control portion **900** controls the exhaust fan **47** to be turned on and off such that the temperature in the casing **3010** is kept in a predetermined range (for example, 80° C. to 90° C.) during the image forming job. The control portion **900** starts (turns on) the operation of the exhaust fan **47** to lower the temperature in the casing **3010** to a predetermined range when the temperature in the casing **3010** is higher than the predetermined range, and stops (turns off) the exhaust fan **47** to raise the temperature in the casing **3010** to the predetermined range when the temperature in the casing **3010** is lower than the predetermined range. The control portion **900** operates the exhaust fan **47** with the air volume as a first air volume during the image forming job. In addition, the control portion **900** also operates the outside air fans **33***a* to **33***c*, the blowing fans **30***a* to **30***c*, and the intake fans **27**, **28**, and **29** at a predetermined air volume during the image forming job.

[0053] Further, the door lock portions **131**, **141**, and **151** and the operation unit **200** are connected to the control portion **900**. The control portion **900** transmits information for changing the display content displayed on the operation unit **200** or receives various types of information, data, and the like input from the operation unit **200** by the user. In a case where the control portion **900** receives an instruction to unlock the door from the operation unit **200**, the control portion **900** controls the door lock portions **131**, **141**, and **151** to unlock the left front door **14**, the right front door **15**, and the upper door **13** (see FIG. **2**A). However, as will be described later, in the present embodiment, the door is not unlocked unless the temperature in the casing **3010** is equal to or lower than a predetermined temperature.

[0054] As described above, in the drying module **3000**, the ink is dried by blowing hot air having a high temperature to the recording material S. In addition, the heating belt **19** is heated by the heating rollers **22** and **23**. Then, during the image forming job, the temperature in the casing **3010** is kept at a high temperature (for example, in a range of 80° C. to 90° C.). As already described, in the related art, the door lock portions **131**, **141**, and **151** are immediately unlocked in response to the user's instruction to unlock the door, and the user can open the upper door **13**, the left front door **14**, and the right front door **15** (see FIG. **2**A). However, when the door is unlocked immediately in response to the user's instruction to unlock the door and the user is ready to open the door, since the user opens the door without knowing that the temperature in the casing **3010** is high, there is a risk that the user will be exposed to high temperatures and may also touch various apparatuses inside the casing **3010** that have become hot. Therefore, it is preferable that the door is unlocked in a case where the temperature in the casing **3010** is equal to or lower than a predetermined temperature. The predetermined temperature may be a temperature (for example, 25° C.) at which the user does not get burned. In addition, in a case where the user performs the maintenance, when it takes time for the temperature in the casing **3010** to become equal to or lower than the predetermined temperature, the image formation cannot be performed, and the operation efficiency of the ink jet recording apparatus **100** is lowered, which is not preferable.

[0055] In the present embodiment, in light of the above, when the door is opened in response to the user's instruction to unlock the door, the temperature in the casing **3010** of the drying module **3000** is quickly lowered, and then the door is unlocked to be opened. In order to quickly lower the temperature in the casing **3010**, the exhaust fan **47** quickly dissipates heat from the inside to the outside of the casing **3010**. Hereinafter, door unlocking processing for achieving the above will be described by using FIG. **6** while referring to FIGS. **3** and **5**. The control portion **900** disables the user from issuing an instruction to unlock the door from the operation unit **200** during the image formation.

Door Unlocking Processing

[0056] As shown in FIG. **6**, the control portion **900** determines whether an instruction to unlock the door is received (step S**1**). When an instruction to unlock the door is not received (NO in step S**1**), the control portion **900** waits for the processing. As described above, an instruction to unlock the door is issued by the user operating the operation unit **200**, and the control portion **900** waits without unlocking the door unless the instruction to unlock the door is acquired.

[0057] When the control portion **900** receives an instruction to unlock the door (YES in step S**1**), the control portion **900** stops the heating of the air taken into the duct **181***a* by the blowing fans **30***a* to **30***c* by the outside air heating heaters **31***a* to **31***c*, and stops the heating of the heating rollers **22** and **23** by the roller heaters **61** and **62** (step S**2**). In addition, the control portion **900** determines whether the temperature (internal temperature) in the casing **3010** is higher than the predetermined temperature based on the detection results of the temperature sensors **25** and **26** (step S**3**). When the temperature in the casing **3010** is equal to or lower than the predetermined temperature (NO in step S**3**), the control portion **900** jumps to the processing of step S**6**.

[0058] When the temperature in the casing **3010** is higher than the predetermined temperature (YES in step S**3**), the control portion **900** drives the exhaust fan **47** (step S**4**). At this time, the control portion **900** starts the operation of the exhaust fan **47** when the exhaust fan **47** is stopped, and continues the operation of the exhaust fan **47** when the exhaust fan **47** is in operation. In addition, it is preferable that the control portion **900** increases the air volume of each fan compared to the air volume of each fan at the time of image formation. For example, it is preferable that the control portion **900** increases the air volume of the blowing fans **30***a* to **30***c* and the exhaust fan **47** to a second air volume larger than a first air volume during image formation. Then, the control portion **900** determines whether the temperature in the casing **3010** is equal to or lower than the predetermined temperature (for example, 25° C.) (step S**5**). The control portion **900** continues to discharge the air in the casing **3010** by operating the exhaust fan **47** in addition to the outside air fans **33***a* to **33***c*, the blowing fans **30***a* to **30***c*, and the intake fans **27**, **28**, and **29** until the temperature in the casing **3010** reaches the predetermined temperature or lower (NO in step S**5**).

[0059] In the present embodiment, the operations of the outside air fans **33***a* to **33***c*, the blowing fans **30***a* to **30***c*, and the intake fans **27**, **28**, and **29** are continued, and the rotation of the heating belt **19** is continued. As a result, the air (outside air) taken in from the outside of the casing **3010** is blown to cool the heating belt **19** without being heated by the outside air heating heaters **31***a* to **31***c*. In addition, the air (outside air) taken in from the outside of the casing **3010** cools not only the heating belt **19** but also each part of the drying belt unit **3300** and the hot air blowing unit **3400**. In addition, since the exhaust fan **47** promotes the discharge of air from the inside to the outside of the casing **3010**, heat can be quickly dissipated from the inside to the outside of the casing **3010**, and the temperature in the casing **3010** can be quickly lowered.

[0060] When the control portion **900** receives an instruction to unlock the door, it is preferable that the control portion **900** drives the outside air fans **33***a* to **33***c*, the blowing fans **30***a* to **30***c*, the intake fans **27**, **28**, and **29**, and the exhaust fan **47** with a predetermined control amount such that the air volume thereof is increased as compared with the air volume during image formation. The predetermined control amount is, for example, a duty ratio of a PWM signal that controls the supply power of each fan. The duty ratio is a predetermined value and may be, for example, 50% or 100%.

[0061] When the temperature in the casing **3010** is lowered to the predetermined temperature or lower (YES in step S**5**), the control portion **900** stops the exhaust fan **47** (step S**6**). In addition, the control portion **900** stops the outside air fans **33***a* to **33***c*, the blowing fans **30***a* to **30***c*, and the intake fans **27**, **28**, and **29**, and further stops the drive roller **20** (step S**6**). After the above-described fans are stopped, the control portion **900** controls the door lock portions **131**, **141**, and **151** to unlock the left front door **14**, the right front door **15**, and the upper door **13** (see FIG. **2**A) (step S**7**).

[0062] As described above, in the present embodiment, in a case where an instruction to unlock the door is issued, the temperature in the casing **3010** is lowered to a predetermined temperature or lower that is safe for the user by the exhaust fan **47**. Then, the door is unlocked by the door lock portions **131**, **141**, and **151**, and the door can be opened. As described above, in the present embodiment, since the user cannot immediately open the door in response to the instruction to unlock the door until the temperature in the casing **3010** is lowered to the predetermined temperature or lower, the door will not be opened while the inside of the casing **3010** is at a high temperature.

[0063] In addition, since the exhaust fan **47** forcibly discharges the air in the casing **3010** to the outside of the casing **3010** by using the air (outside air) blown from the blowing fans **30***a* to **30***c* to the heating belt **19**, the time required for the temperature in the casing **3010** to be lowered to a predetermined temperature or lower can be shortened. As a result, the present invention contributes to improvement of the operation efficiency of the ink jet recording apparatus **100**.

[0064] In the present embodiment, the temperature sensors **25** and **26** detect the temperature of the heating belt **19**, and the door is unlocked when the temperature of the heating belt **19** is equal to or lower than a predetermined temperature. This is because, in a case where the user accesses the inside of the casing, the user may touch the heating belt **19** having a high temperature. Since the heating belt **19** becomes hot and the user is also likely to touch the heating belt **19**, the temperature sensors **25** and **26** that detect the temperature of the heating belt **19** are used as a trigger for unlocking.

Second Embodiment

[0065] Next, a second embodiment will be described. As described above, in a case where a jam occurs in which the recording material S is clogged in the middle of being conveyed, the user can remove the recording material S by opening the upper door **13**, the left front door **14**, and the right front door **15**. Even in a case where a jam occurs, when the door is unlocked immediately in response to the user's instruction to unlock the door, there is a risk that the user will open the door without knowing that the temperature in the casing **3010** is high. Therefore, even in a case where a jam occurs, it is preferable that the door is unlocked in a case where the temperature in the casing **3010** is equal to or lower than the predetermined temperature. Hereinafter, door unlocking processing in a case where a jam occurs will be described by using FIG. **7** while referring to FIGS. **3** and **5**.

[0066] The control portion **900** determines whether a jam has occurred in which the recording material S is clogged in the middle of being conveyed, based on the detection results of the recording material detection sensors **45** and **46** (step S**11**). When a jam occurs (YES in step S**11**), the control portion **900** controls the drive roller **20** to stop the rotation to stop the conveyance of the recording material S by the heating belt **19**, and controls the outside air heating heaters **31***a* to **31***c* and the roller heaters **61** and **62** to stop the heating of the heating belt **19** (step S**12**). Then, the control portion **900** drives the exhaust fan **47** (step S**13**). In this case, the control portion **900** starts the operation of the exhaust fan **47** when the exhaust fan **47** is stopped, and continues the operation of the exhaust fan **47** when the exhaust fan **47** is in operation. In addition, it is preferable that the control portion **900** increases the air volume of each fan compared to the air volume of each fan during image formation. For example, it is preferable that the control portion **900** increases the air volume of the blowing fans **30***a* to **30***c* and the exhaust fan **47** to a second air volume larger than a first air volume during image formation.

[0067] The control portion **900** determines whether an instruction to unlock the door is received (step S**14**). When an instruction to unlock the door is not acquired (NO in step S**14**), the control portion **900** determines whether a predetermined time has elapsed from the detection of a jam (step S**19**). When a predetermined time has not elapsed from the detection of a jam (NO in step S**19**), the control portion **900** returns to the processing of step S**14**. When a predetermined time has elapsed from the detection of a jam (YES in step S**19**), the control portion **900** stops the outside air fans **33***a* to **33***c*, the blowing fans **30***a* to **30***c*, the intake fans **27**, **28**, and **29**, and the exhaust fan **47** (step S**20**).

[0068] In this case, since the heating by the outside air heating heaters **31***a* to **31***c* and the roller heaters **61** and **62** is stopped and the exhaust fan **47** is driven, the temperature in the casing **3010** is lowered with the passage of time. Then, after a predetermined time elapses from the detection of a jam, the temperature in the casing **3010** is lowered to a predetermined temperature or lower that is safe for the user. The predetermined time here is a time sufficient for the temperature in the casing **3010** to drop to the predetermined temperature or lower, and is stored in advance in the ROM or the like. In this case, since an instruction to unlock the door is not acquired, the control portion **900** does not unlock the door. However, in a case where the user issues an instruction to unlock the door later, the temperature in the casing **3010** has already been lowered to a predetermined temperature or lower, and thus the door is immediately unlocked.

[0069] Meanwhile, when an instruction to unlock the door is received (YES in step S**14**), the control portion **900** increases the air volume of each fan as compared with the air volume of each fan during image formation (step S**15**). For example, the control portion **900** increases the air volume of the blowing fans **30***a* to **30***c* and the exhaust fan **47** to a second air volume larger than a first air volume during image formation. Then, the control portion **900** determines whether the temperature in the casing **3010** is equal to or lower than the predetermined temperature (for example, 25° C.) (step S**16**). The control portion **900** operates the exhaust fan **47** in addition to the outside air fans **33***a* to **33***c*, the blowing fans **30***a* to **30***c*, and the intake fans **27**, **28**, and **29** until the temperature in the casing **3010** reaches the predetermined temperature or lower, and continues to discharge the air in the casing **3010** (NO in step S**16**). When the temperature in the casing **3010** is lowered to the predetermined temperature or lower (YES in step S**16**), the control portion **900** stops the outside air fans **33***a* to **33***c*, the blowing fans **30***a* to **30***c*, the intake fans **27**, **28**, and **29**, and the exhaust fan **47** (step S**17**). After the above-described fans are stopped, the control portion **900** controls the door lock portions **131**, **141**, and **151** to unlock the left front door **14**, the right front door **15**, and the upper door **13** (see FIG. **2**A) (step S**18**).

[0070] As described above, in a case where a jam occurs, the exhaust fan **47** is driven after the heating by the outside air heating heaters **31***a* to **31***c* and the roller heaters **61** and **62** is stopped. In a case where an instruction to unlock the door is issued, the door is not unlocked by the door lock portions **131**, **141**, and **151** unless the temperature in the casing **3010** is lowered to a predetermined temperature or lower that is safe for the user by the exhaust fan **47**. Therefore, even in a case where a jam occurs, the user cannot immediately open the door in response to the instruction to unlock the door until the temperature in the casing **3010** is lowered to the predetermined temperature or lower, and thus the door will not be opened while the inside of the casing **3010** is at a high temperature.

OTHER EMBODIMENTS

[0071] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit

(CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0072] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0073] This application claims the benefit of Japanese Patent Application No. 2024-035942, filed Mar. 8, 2024, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. An ink jet recording apparatus forming an image on a recording material by jetting an ink, the apparatus comprising:

a conveyance portion configured to convey the recording material on which the image is formed by the ink;

a drying portion including a heater and a blowing fan blowing air toward the conveyance portion and configured to dry the ink by blowing air heated by the heater to the recording material conveyed by the conveyance portion;

a casing configured to accommodate the conveyance portion;

a temperature sensor configured to detect a temperature in the casing;

an exhaust fan configured to exhaust air in the casing to an outside of the casing and to be controlled to an operation state and a stop state based on a detection result of the temperature sensor;

a door provided to be opened and closed on the casing to allow access to the conveyance portion;

a lock mechanism configured to lock the door to the casing in a closed state; and

a control portion configured to control the drying portion, the lock mechanism, and the exhaust fan,

wherein, in a case where the control portion receives an instruction to unlock the door locked by the lock mechanism, the control portion is configured to stop heating of the air by the heater,

start an operation of the exhaust fan when the exhaust fan is stopped,

continue the operation of the exhaust fan when the exhaust fan is in operation, and

unlock the door locked by the lock mechanism after the temperature in the casing reaches a predetermined temperature or lower.

2. The ink jet recording apparatus according to claim **1**, wherein

the exhaust fan is operable at a first air volume and a second air volume larger than the first air volume, and

in a case where the control portion receives the instruction to unlock the door, the control portion is configured to start the operation of the exhaust fan at the second air volume when the exhaust fan is stopped, and change an air volume of the exhaust fan to the second air volume when the exhaust fan is in operation at the first air volume.

3. The ink jet recording apparatus according to claim **1**, wherein

the blowing fan is operable at a first air volume and a second air volume larger than the first air volume, and

in a case where the control portion receives the instruction to unlock the door, the control portion is configured to change an air volume of the blowing fan from the first air volume to the second air volume.

4. The ink jet recording apparatus according to claim **1**, wherein

the control portion is configured to operate the exhaust fan and the blowing fan until the temperature in the casing reaches the predetermined temperature or lower.

5. The ink jet recording apparatus according to claim **1**, wherein

the door is an upper door that is provided above the casing to be opened and closed in an up-down direction with respect to the casing, and

the drying portion is disposed on the upper door and moves together with the upper door to be accessible from above to the conveyance portion accommodated in the casing.

6. The ink jet recording apparatus according to claim **1**, wherein

the door is a front door that is provided to be opened and closed in a direction intersecting an up-down direction with respect to the casing, and

the conveyance portion is in an accessible state in a state in which the front door is opened.

7. An ink jet recording apparatus forming an image on a recording material by jetting an ink, the apparatus comprising:

a conveyance portion configured to convey the recording material on which the image is formed by the ink;

a detection unit configured to detect the recording material conveyed by the conveyance portion;

a drying portion including a heater and a blowing fan blowing air toward the conveyance portion and configured to dry the ink by blowing air heated by the heater to the recording material conveyed by the conveyance portion;

a casing configured to accommodate the conveyance portion;

a temperature sensor configured to detect a temperature in the casing;

an exhaust fan configured to exhaust air in the casing to an outside of the casing and to be controlled to an operation state and a stop state based on a detection result of the temperature sensor;

a door that is provided to be opened and closed on the casing to allow access to the conveyance portion;

a lock mechanism configured to lock the door to the casing in a closed state; and

a control portion configured to control the drying portion, the lock mechanism, and the exhaust fan,

wherein, in a case where an error occurs in a detection result of the detection unit, the control portion is configured to stop heating of the air by the heater, and start the operation of the exhaust fan when the exhaust fan is stopped, and

wherein when the control portion receives an instruction to unlock the door locked by the lock mechanism after the error occurs, the control portion is configured to unlock the door locked by the lock mechanism after the temperature in the casing reaches a predetermined temperature or lower.

8. The ink jet recording apparatus according to claim **7**, wherein

the exhaust fan is operable at a first air volume and a second air volume larger than the first air volume, and

in a case where the control portion receives the instruction to unlock the door, the control portion is configured to change an air volume of the exhaust fan from the first air volume to the second air volume.

9. The ink jet recording apparatus according to claim **7**, wherein

the blowing fan is operable at a first air volume and a second air volume larger than the first air volume, and

in a case where the control portion receives the instruction to unlock the door, the control portion is configured to change an air volume of the blowing fan from the first air volume to the second air volume.

10. The ink jet recording apparatus according to claim **7**, wherein

the control portion is configured to operate the exhaust fan and the blowing fan until the temperature in the casing reaches the predetermined temperature or lower.

11. The ink jet recording apparatus according to claim **7**, wherein

the door is an upper door that is provided above the casing to be opened and closed in an up-down direction with respect to the casing, and

the drying portion is disposed on the upper door and moves together with the upper door to be accessible from above to the conveyance portion accommodated in the casing.

12. The ink jet recording apparatus according to claim **7**, wherein

the door is a front door that is provided to be opened and closed in a direction intersecting an up-down direction with respect to the casing, and

the conveyance portion is in an accessible state in a state in which the front door is opened.

13. The ink jet recording apparatus according to claim **1**, wherein

the conveyance portion includes a belt that rotates to convey the recording material, and a heating portion that heats the belt, and

the temperature sensor is configured to detect a temperature of the belt.

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